

NOVI LIGURE, Italy

WMO: 161180

Lat: 44.767N

Lon: 8.783E

Elev: 614

StdP: 14.37

Time Zone: 1.00 (EUC)

Period: 86-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	24.5	8.2	8.6	43.3	14.2	11.6	41.5	19.9	43.6	16.2	42.9	2.9	140	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.7	88.5	67.8	86.4	67.8	84.4	66.6	73.7	83.4	72.3	81.3	70.8	79.4	5.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	113.8	77.7	69.5	111.1	77.4	67.9	104.8	76.3	37.6	83.8	36.2	81.0	35.0	79.4	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	11.8	9.7	DB	18.1	91.5	4.3	1.7	15.1	92.7	12.6	93.7	10.2	94.6	7.1	95.8	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.1	35.7	40.5	48.0	52.9	63.1	68.1	73.5	74.2	66.7	56.5	44.3	36.8	(6)	
	DBStd	14.50	4.96	6.99	6.09	4.86	5.76	4.95	4.22	4.23	4.48	6.15	5.22	5.69	(7)	
	HDD50	1460	444	272	110	25	0	0	0	0	19	178	410		(8)	
	HDD65	4410	909	686	528	362	102	25	1	33	269	620	874		(9)	
	CDD50	3325	0	6	47	113	407	543	730	750	502	219	8	0	(10)	
	CDD65	801	0	0	0	0	42	118	266	286	85	4	0	0	(11)	
	CDH74	6245	0	0	1	2	281	791	2278	2372	495	25	0	0	(12)	
	CDH80	1800	0	0	0	0	34	156	732	808	69	1	0	0	(13)	
(14) Wind	WSAvg	3.1	2.1	2.8	3.9	4.3	4.0	3.9	3.7	3.5	2.8	2.4	2.0	2.2	(14)	
(15) Precipitation	PrecAvg	33.0	2.6	2.3	1.9	2.8	2.2	2.0	1.2	2.2	3.9	4.6	5.0	2.6	(15)	
	PrecMax	52.3	10.7	6.0	6.2	7.2	6.4	5.0	3.2	4.8	11.4	10.3	16.5	7.0	(16)	
	PrecMin	20.3	0.1	0.1	0.4	0.2	0.0	0.0	0.1	0.1	0.2	1.0	0.2	0.0	(17)	
	PrecStd	9.1	2.6	1.8	1.5	2.0	1.4	1.4	0.8	1.4	3.1	2.6	4.1	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	63.9	71.8	73.1	83.7	87.6	91.1	91.4	85.5	77.2	59.4	55.0	(19)	
		MCWB	43.5	51.4	55.2	58.8	66.2	69.8	68.7	62.8	66.0	59.4	51.2	45.7	(20)	
	2%	DB	48.1	58.7	66.4	69.2	80.4	84.5	88.7	89.1	82.0	71.8	56.3	49.6	(21)	
		MCWB	43.2	49.3	52.2	55.5	64.1	67.3	68.7	66.0	66.4	59.9	50.6	45.1	(22)	
	5%	DB	45.0	54.9	62.7	66.0	77.0	82.0	86.4	87.2	79.1	68.0	54.0	46.6	(23)	
		MCWB	42.1	47.0	50.7	54.2	61.3	65.2	68.3	67.4	64.2	57.8	49.5	43.7	(24)	
	10%	DB	43.0	51.6	59.2	62.7	74.1	78.9	84.3	84.7	76.8	64.6	52.0	44.7	(25)	
		MCWB	40.9	46.0	49.5	52.5	60.0	63.4	66.8	66.3	62.9	57.3	48.6	42.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	53.4	57.0	60.0	69.9	74.4	75.7	75.3	73.1	64.5	55.0	50.9	(27)	
		MCDB	49.1	64.4	68.5	69.4	82.0	85.1	84.8	85.5	80.5	71.2	58.1	52.7	(28)	
	2%	WB	45.7	50.6	54.6	57.3	67.2	70.9	73.6	73.7	70.4	62.8	53.4	47.6	(29)	
		MCDB	46.9	57.3	65.1	65.3	76.8	80.5	83.2	83.6	77.8	68.1	55.9	49.3	(30)	
	5%	WB	43.9	49.5	52.7	55.5	64.4	68.7	72.3	72.3	68.2	61.4	51.2	45.9	(31)	
		MCDB	45.6	54.7	62.1	61.8	73.1	77.6	81.5	81.2	74.9	65.6	53.5	47.3	(32)	
	10%	WB	42.1	48.1	51.2	54.1	62.3	66.9	70.8	70.8	66.5	59.8	50.2	43.9	(33)	
		MCDB	43.7	51.9	58.2	61.0	70.2	75.1	79.7	78.8	72.0	63.2	52.0	45.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	15.1	18.6	16.0	18.0	17.7	19.6	19.7	18.0	13.1	11.2	11.0	(35)	
		MCDBR	15.6	21.4	24.4	22.1	22.1	20.7	22.0	22.2	20.9	18.9	15.6	15.1	(36)	
	5% WB	MCWBR	10.4	11.8	11.8	10.6	8.1	7.1	7.4	7.0	7.4	7.9	8.9	10.7	(37)	
		MCWBR	11.4	20.7	22.9	16.4	18.6	17.6	18.4	18.3	17.2	13.7	8.3	9.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.336	0.359	0.403	0.444	0.443	0.462	0.448	0.441	0.430	0.408	0.364	0.332	(40)	
		taud	2.415	2.330	2.217	2.121	2.175	2.167	2.212	2.241	2.239	2.269	2.360	2.443	(41)	
	Ebn at Noon		245	259	261	259	264	258	260	258	249	236	229	234	(42)	
		Edh at Noon	24	31	39	47	46	46	44	41	38	32	25	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		475	720	1139	1470	1822	2096	2157	1833	1372	804	482	399	(44)	
	RadStd		62	106	150	145	148	112	118	108	90	117	72	56	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (43 neighbors)		+0.65	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	+139	+41	

Nomenclature: See separate page